

Effect of cruciferous vegetable consumption on heterocyclic aromatic amine metabolism in man.

.

The consumption of cooked meat appears to predispose individuals to colonic cancer and heterocyclic aromatic amines (HA), formed during the cooking of meat, have been suggested as aetiological agents. Consumption of cruciferous vegetables is thought to protect against cancer. To study the effect of cruciferous vegetables on heterocyclic aromatic amine metabolism in man, a three-period, dietary intervention study has been carried out with 20 non-smoking Caucasian male subjects consuming cooked meat meals containing known amounts of these carcinogens. A high cruciferous vegetable diet (250 g each of Brussels sprouts and broccoli per day) was maintained during period 2 but such vegetables were excluded from periods 1 and 3. At the end of each period, subjects consumed a cooked meat meal and urinary excretion of the HA 2-amino-3,8-dimethylimidazo(4,5-f)quinoxaline (MeIQx) and 2-amino-1-methyl-6-phenylimidazo(4,5-b)pyridine (PhIP) was measured. Following a 12 day period of cruciferous vegetable consumption (period 2), induction of hepatic CYP1A2 activity was apparent from changes in the kinetics of caffeine metabolism. Excretion of MeIQx and PhIP in urine at the end of this period of the study was reduced by 23 and 21%, respectively, compared with period 1. This reduction in excretion is probably due to an increase in amine metabolism that might be expected given the observed increase in CYP1A2 activity, since this enzyme has been shown to be primarily responsible for the oxidative activation of MeIQx and PhIP in man. In period 2, urinary mutagenicity was increased relative to period 1 by 52 and 64% in the absence and presence, respectively, of a human liver microsomal activation system, yet no evidence was found of PhIP adduction to lymphocyte DNA, a potential biomarker of the activation process. After another 12 days without cruciferous vegetables (period 3 of the study), the kinetics of caffeine metabolism had returned to original values but excretion of MeIQx and PhIP was still reduced by 17 and 30%, respectively, and urinary mutagenicity (with metabolic activation) was still elevated compared with period 1. This prolonged response of amine metabolism to the cruciferous vegetable diet, shown especially with PhIP, suggests that enzyme systems other than CYP1A2 are involved and affected by a cruciferous vegetable diet.